Experiment 7: Breadth First Search (BFS)

Aim

To write a Python program to implement Breadth First Search (BFS) traversal on a graph.

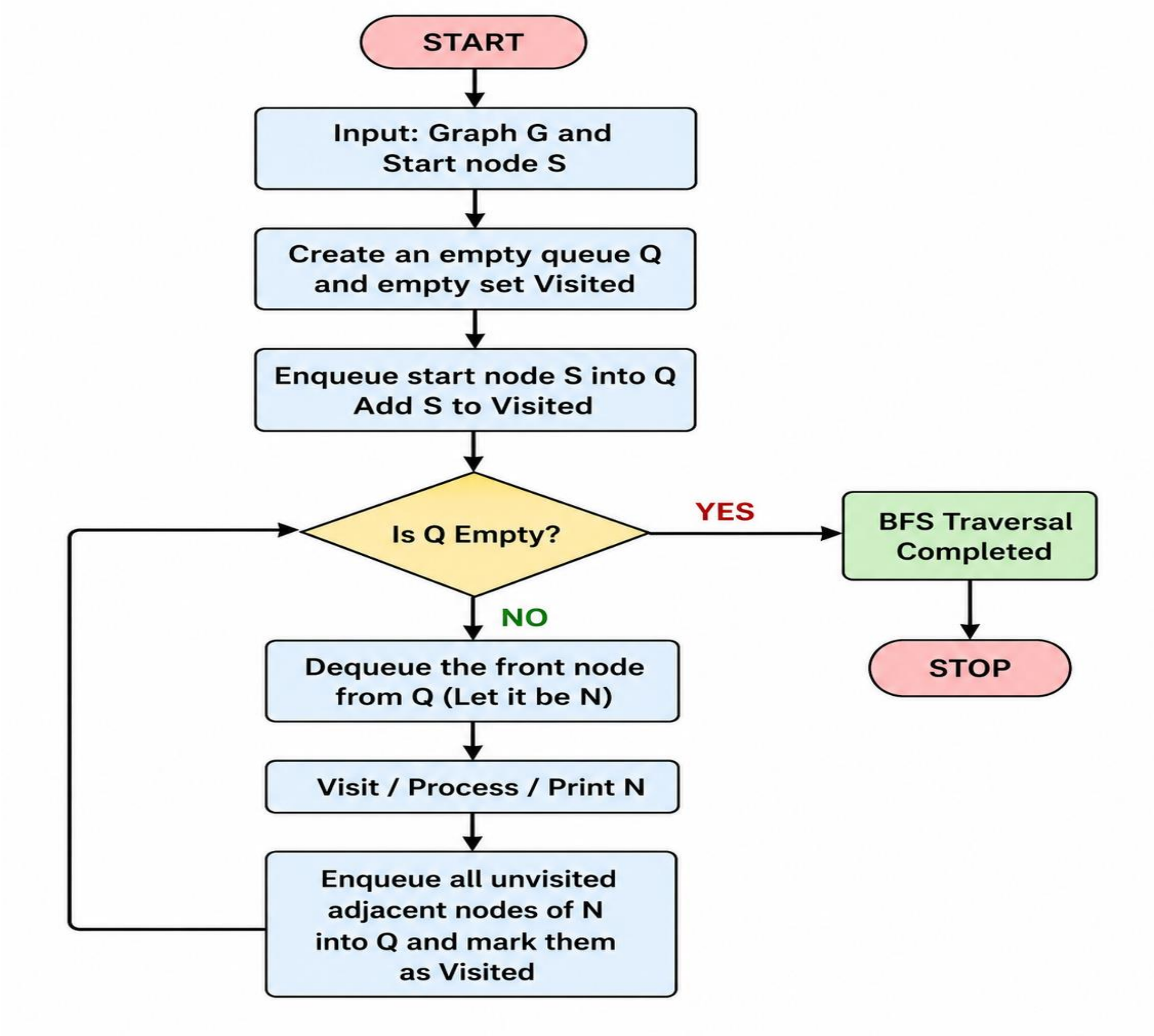
Objective

To traverse all the nodes of a graph level by level using the BFS algorithm and understand the working of queue-based graph traversal.

Algorithm

- Create a graph using an adjacency list.
- Initialize a queue with the starting node.
- Mark the starting node as visited.
- Remove a node from the queue.
- Print the node.
- Add all unvisited adjacent nodes to the queue.
- Repeat steps 4–6 until the queue becomes empty.

FLOWCHART:



Code

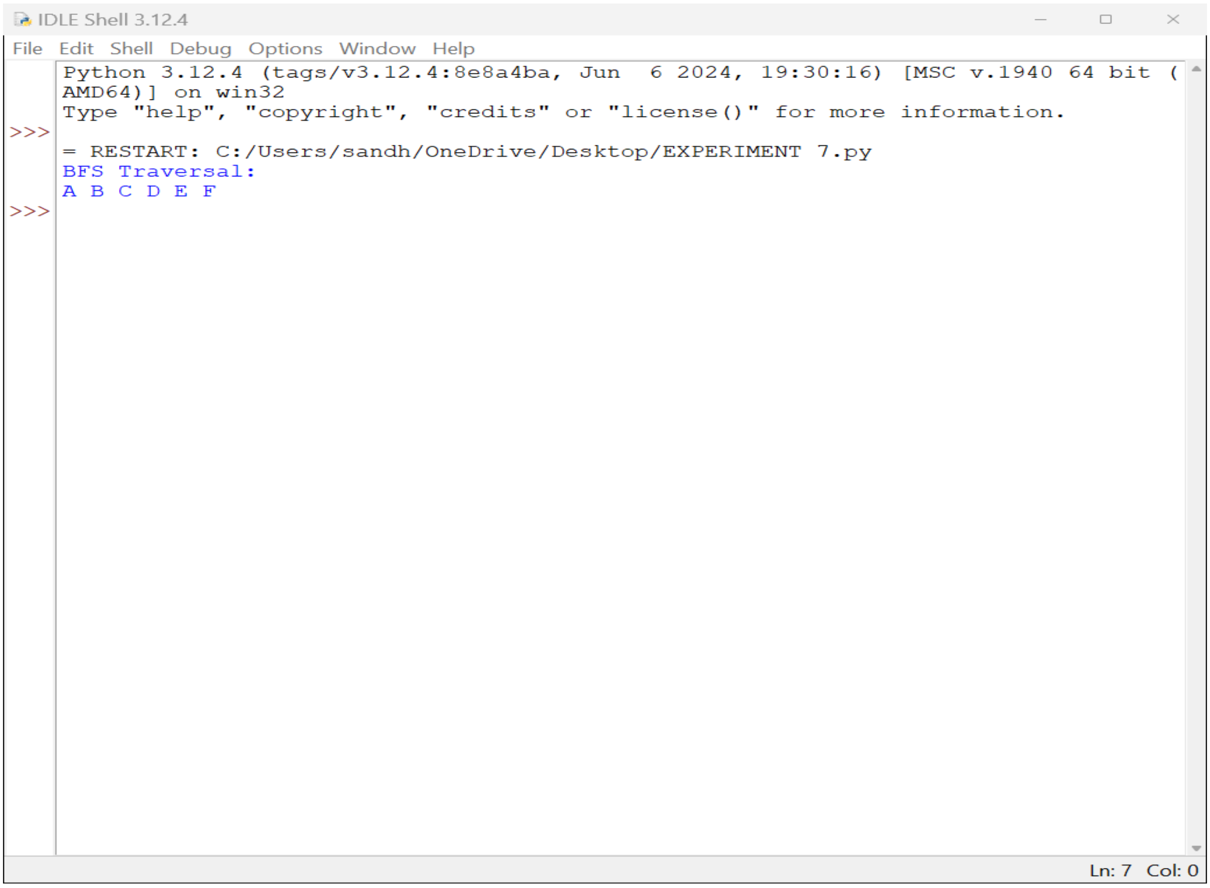
Python

Run

```
from collections import deque


g = {'A':['B','C'], 'B':['D'], 'C':['E'], 'D':[], 'E':[]}


q = deque(['A'])

v = set(['A'])


print("BFS Traversal:")
```

```
while q:
   n = q.popleft()
   print(n, end=' ')
   for i in g[n]:
      if i not in v:
         v.add(i)
         q.append(i)
```

Output



Result

The BFS traversal of the given graph was successfully performed using Python, and the nodes were visited in the order: **A → B → C → D → E**.

Conclusion

The Breadth First Search algorithm was implemented successfully in Python. BFS visits graph nodes level by level using a queue and is useful for shortest-path and graph traversal applications.

ChatGPT can make mistakes. Check important info.